

FAgentLLM: Five Agents, One Vision: Smarter Finance, Better Decisions

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Abstract – Enterprise finance faces a paradox of automation: although ERP and RPA systems accelerate operations, systemic intelligence is weakened by siloed data and fragmented decision-making. This paper introduces **FAgentLLM**, a unified multi-agent architecture designed to overcome these high-performing silos by enabling coordinated, cross-domain financial intelligence. Through a structured review of literature (2022–2025), the study examines five core financial domains—invoice processing, budgeting, reconciliation, credit assessment, and cash management—and traces the evolution from rule-based automation to LLM-orchestrated Multi-Agent Systems (MAS). Existing AI models demonstrate strong local performance (e.g., near-perfect forecasting accuracy) yet fail to capture causal and contextual interdependencies across functions. FAgentLLM addresses this limitation by employing Large Language Models as a cognitive orchestration layer that integrates unstructured semantic reasoning with mathematically grounded execution across agents. The proposed architecture restores context-aware judgment to automated finance, enabling explainable, adaptive, and system-level financial intelligence.

Keywords - Large Language Models (LLMs); Multi-Agent Systems (MAS); Financial Automation; FAgentLLM; Cross-Domain Coordination; Explainable AI (XAI); Systemic Intelligence; Enterprise Resource Planning (ERP).

I INTRODUCTION

Enterprise finance is no longer limited by a lack of data; it is constrained by how poorly that data is processed, connected, and acted upon. Core financial operations such as invoice processing, budgeting, reconciliation, credit tracking, and cash management remain heavily manual, fragmented across systems, and reactive by design. Finance teams spend a disproportionate amount of time validating entries, resolving mismatches, and chasing approvals, while strategic analysis and proactive decision-making are delayed or deprioritized.

Over the past decade, organizations have attempted to address these inefficiencies through Enterprise Resource Planning (ERP) extensions and Robotic Process Automation (RPA). While such solutions reduce surface-level repetition, they fundamentally operate on static rules and predefined workflows. As a result, they fail when confronted with unstructured financial documents, evolving compliance requirements, cross-departmental dependencies, and context-sensitive decision logic. In practice, this has led to a paradox: financial systems are increasingly automated, yet operational friction and oversight costs continue to rise.

Recent advances in *Large Language Models (LLMs)* and *agentic artificial intelligence* introduce a decisive shift in how financial automation can be conceived. Unlike traditional automation tools, LLM-powered agents are capable of interpreting unstructured financial data, reasoning over business context, and coordinating actions across interconnected workflows. This enables automation not merely at the task level, but at the process and decision level, where intelligence, adaptability, and explainability are critical.

This literature review critically examines existing research and systems related to intelligent financial automation, with particular focus on five foundational enterprise finance domains: *invoice management*, *budget management*, *reconciliation*, *credit tracking and collection*, and *cash management*. Rather than evaluating these domains in isolation, the review emphasizes their structural interdependencies and the limitations of current siloed solutions. The analysis establishes the motivation for **FAgentLLM**, a unified architecture of five LLM-powered financial agents designed to transform fragmented financial operations into a coordinated, explainable, and decision-oriented automation ecosystem.

II THEORETICAL BACKGROUND

This section sets the lens for reading the literature that follows. Rather than listing systems or methods, it interrogates the assumptions embedded in claims of intelligent financial automation; what counts as intelligence, how coordination is achieved, and where automation ends and judgment is presumed to begin. Making these assumptions explicit is necessary to distinguish what prior work genuinely establishes from what it merely suggests.

2.1 Conceptual Limitations of Traditional Financial Automation

Financial automation has achieved remarkable operational efficiency, yet it has simultaneously failed to deliver systemic financial intelligence. Invoices are processed faster than ever, transactions clear in milliseconds, and reports are generated automatically. By conventional operational metrics, modern financial systems represent a technological triumph. Yet when senior financial decision-makers are asked whether their organizations truly understand their financial position in real time, whether decisions propagate intelligently across departments, or whether exceptions are handled contextually rather than escalated to human intervention, the answer is consistently negative.

This gap does not originate from a lack of technology. Rather, it reflects a conceptual misalignment between what has been automated and what constitutes meaningful financial intelligence.

Prior to large-scale automation, financial operations were slow, manual, and error-prone, yet inherently context-aware. An accountant reviewing an invoice did not merely verify numerical correctness. The decision implicitly incorporated vendor history, budget availability, liquidity conditions, regulatory obligations, and broader strategic priorities. Judgment emerged from the simultaneous consideration of multiple interdependent factors. The intelligence of the system resided in the human mind, which is inherently relational: capable of connecting signals, weighing trade-offs, and adapting to change.

Enterprise Resource Planning (ERP) systems introduced in the 1990s promised to digitize this intelligence. In practice, they digitized structure rather than judgment. Financial operations were decomposed into schemas—entities, relationships, and predefined workflows—which proved highly effective under stable and predictable conditions. However, when operational reality deviated from the encoded model—through changing vendor formats, evolving regulations, or organizational restructuring—the system interpreted deviation as error rather than context.

ERP systems did not eliminate judgment; they externalized it, requiring humans to continuously mediate between rigid system assumptions and fluid financial reality.

Robotic Process Automation (RPA) emerged in the 2010s to automate what ERP systems could not absorb: the residual manual actions that bridged incompatible systems. While often presented as intelligent automation, RPA fundamentally operates as behavioral imitation.

Scripts replicate recorded actions with precision, yet fail when interfaces or conditions change. Where human operators adapt, RPA halts. It delivers execution without understanding, repetition without reflection—automation in its most literal and constrained form.

The current wave of intelligent automation integrates machine learning and natural language processing to extend beyond execution. These systems introduce meaningful capabilities, including document classification, anomaly detection, and financial forecasting. However, closer examination reveals a persistent limitation. Such systems primarily recognize patterns in historical data, optimize local objectives, and respond to correlations rather than causal dependencies. Financial domains remain functionally siloed.

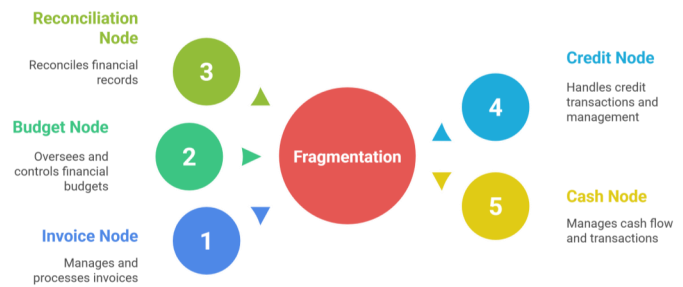


Figure 1 Fragmentation in Financial Systems

An invoice approval system does not reason about near-term liquidity constraints. A budget forecasting model does not assess downstream credit exposure. A reconciliation tool does not interpret recurring mismatches as indicators of systemic process failure. Each system performs its isolated function competently, yet collectively remains blind to the interdependencies that define financial reality.

This reveals a central paradox: as individual financial processes become more efficient, the visibility of their systemic interactions diminishes. Automation has accelerated execution to such an extent that causal relationships collapse into narrow reporting windows.

Despite this compression, systems continue to operate as if financial domains are independent. Speed without coordination does not yield intelligence; it produces sophisticated fragmentation.

This pattern is not incidental but reflects a broader historical trajectory in the design of financial automation.

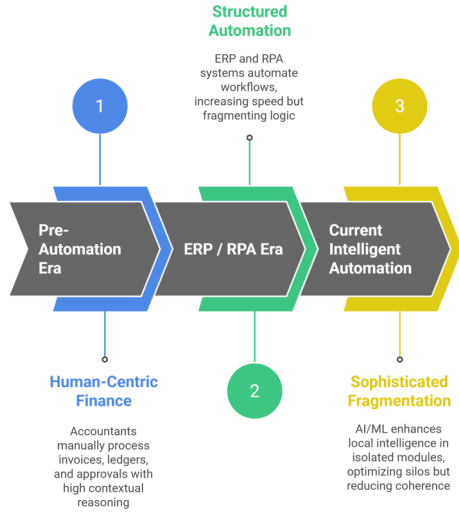


Figure 2 Evolution of Financial Automation

As illustrated in Figure 2, the evolution of financial automation reflects a shift from human-centric judgment to increasingly structured and automated systems. While ERP and RPA technologies enhanced efficiency, they also decomposed financial reasoning into rigid workflows, leading to fragmented intelligence rather than integrated understanding.

The failure is therefore architectural rather than algorithmic. Financial operations are not independent tasks but nodes within a causal network where actions propagate across domains. Pre-automation systems handled this complexity implicitly through human judgment. Automation was expected to formalize and preserve these relationships. Instead, intelligence was atomized into specialized tools that execute effectively in isolation yet fail collectively.

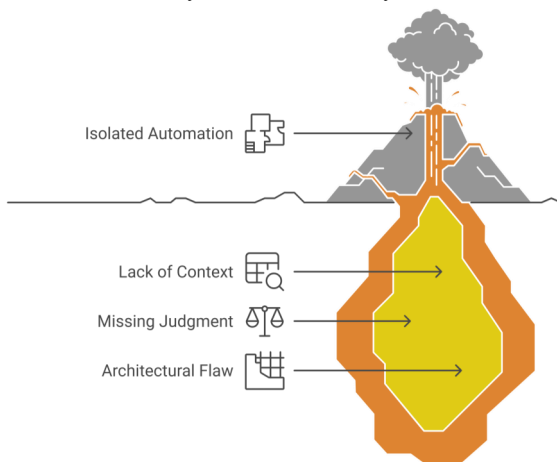


Figure 3 Architectural Failure in Financial Automation.

True financial intelligence requires contextual reasoning, causal awareness, and adaptive coordination. This standard provides the lens through which the following theoretical foundations are examined. The critical question is not whether financial tasks can be automated, but whether automation can preserve and enhance the judgment that made manual processes effective.

2.2 Large Language Models as an Intelligence Layer

Large Language Models (LLMs) are deep neural architectures primarily based on the Transformer model, which employs self-attention mechanisms to model long-range contextual dependencies within data (Vaswani et al., 2017b). Through large-scale pre-training on heterogeneous corpora followed by task-specific fine-tuning, LLMs acquire robust capabilities in semantic understanding, abstraction, and reasoning.

Within enterprise financial systems, LLMs are most appropriately conceptualized not as conversational interfaces, but as an intelligence layer operating above deterministic and statistical components. Financial workflows involve heterogeneous documents, evolving schemas, regulatory constraints, and implicit contextual dependencies that cannot be reliably addressed through static rules alone. LLMs enable interpretation of meaning, detection of semantic inconsistencies, inference of intent, and reasoning over semi-structured financial artifacts such as invoices, reconciliation statements, and compliance records.

At the same time, LLMs are not designed to replace deterministic computation or transactional logic. Their probabilistic nature limits reliability in tasks requiring strict arithmetic precision or guaranteed consistency. This motivates a hybrid architectural paradigm in which LLMs support semantic reasoning and decision guidance, while structured systems retain responsibility for validation, execution, and persistence. This separation of concerns constitutes a foundational design principle of FAgentLLM.

2.3 Intelligent Agents and Multi-Agent Systems

An intelligent agent is an autonomous software entity capable of perceiving its environment, making decisions, and acting to achieve defined objectives (Brewka, 1996b).

When multiple agents interact within a shared environment, the resulting system forms a Multi-Agent System (MAS), characterized by decentralized control, partial observability, and coordinated behavior.

Enterprise financial operations align naturally with this paradigm. Invoice processing, budgeting, reconciliation, credit management, and cash flow monitoring represent distinct operational responsibilities governed by different constraints, data sources, and performance criteria. Modeling these functions as autonomous agents enables modularity, scalability, and fault isolation, while preserving the ability to coordinate decisions across domains.

In FAgentLLM, each financial agent operates within a clearly bounded scope while contributing to a shared financial state. This structure enhances transparency and traceability by enabling decisions to be attributed to specific agents and workflows. Moreover, MAS architectures support incremental evolution, allowing new capabilities to be introduced without destabilizing existing financial processes.

2.4 LLM-Orchestrated Multi-Agent Architectures

Recent advances in artificial intelligence have enabled the integration of LLMs as reasoning and orchestration components within agent-based systems. In LLM-orchestrated MAS architectures, language models guide agent behavior by interpreting goals, selecting tools, coordinating actions, and managing inter-agent communication (Chen et al., 2025b)

This approach differs fundamentally from monolithic LLM systems, where a single model attempts to perform all tasks without structural decomposition. By embedding LLMs within agents rather than centralizing intelligence, the system preserves domain boundaries, enforces governance constraints, and improves explainability. Each agent leverages LLM reasoning to interpret context and propose actions, while execution remains constrained by predefined workflows and validation layers.

Within FAgentLLM, LLMs function as cognitive controllers that enhance each agent's capacity to reason about financial context, resolve ambiguity, and coordinate decisions across domains. Structured message-passing enables workflows in which actions taken by one agent propagate meaningfully to others, ensuring systemic coherence rather than isolated optimization.

2.5 Intelligent Document Processing for Financial Data

Financial workflows rely heavily on documents such as invoices, bank statements, contracts, and regulatory records. Intelligent Document Processing (IDP) integrates Optical Character Recognition (OCR), layout analysis, and semantic reasoning to transform these artifacts into structured, machine-readable representations (Alonso-Rocha et al., 2025b)

Traditional OCR systems focus on text extraction, often failing to capture spatial relationships and semantic meaning. Modern IDP pipelines extend this capability through layout-aware models and LLM-based reasoning, enabling identification of relevant fields, validation of inter-field relationships, and detection of anomalies. This shift from extraction to interpretation is critical for scalable financial automation.

In FAgentLLM, IDP serves as a foundational capability, particularly for the Invoice Management Agent, while also supporting reconciliation and compliance processes. By enabling agents to reason about document content in context, document handling becomes an intelligent decision-making activity rather than a passive ingestion step.

2.6 Financial Workflow Automation and Compliance Theory

Financial workflows are governed by strict internal controls, regulatory obligations, and audit requirements. Automation in this domain must therefore prioritize traceability, explainability, and compliance by design. Unlike general business processes, financial systems cannot tolerate opaque decision logic or uncontrolled automation (Cheng et al., 2025b)

Agent-based workflow automation provides a structured mechanism for enforcing governance. Each agent operates under explicit policies, maintains detailed logs, and supports human-in-the-loop intervention where required. By embedding compliance logic directly within agent responsibilities, automation strengthens rather than weakens financial control.

FAgentLLM operationalizes this principle by integrating intelligent agents within clearly defined workflows that preserve approval hierarchies, validation checkpoints, and reporting obligations. LLM-generated recommendations are contextualized and explainable, enabling finance professionals to understand, audit, and justify automated decisions.

2.7 Supporting Mathematical and Technical Foundations

While semantic reasoning and agent orchestration constitute the conceptual core of FAgentLLM, the system is grounded in a set of mathematical and technical foundations that enable reliable execution across financial workflows. These foundations are intentionally interpretable and explicitly aligned with established financial decision-support practices, in which analytical models inform human judgment while preserving transparency, auditability, and regulatory compliance.

Rather than pursuing opaque or fully autonomous optimization, FAgentLLM adopts lightweight analytical formulations that reflect how financial decisions are made in practice: through traceable inputs, policy-constrained logic, and explainable outputs. This design choice is critical in regulated enterprise environments, where the legitimacy of a system depends not only on predictive performance, but on its ability to justify and defend its recommendations.

2.7.1 Similarity-Based Matching for Reconciliation

Automated reconciliation requires the alignment of internal financial records with corresponding external records obtained from banks, suppliers, or partners. Given the heterogeneity of data formats, reference conventions, and reporting structures, exact matching is frequently infeasible. Reconciliation is therefore modeled as a similarity-based matching problem rather than a strict equality check.

Let x and y denote vector representations of an internal transaction and an external transaction, respectively. These vectors encode relevant attributes such as transaction amount, date, counterparty identifiers, and descriptive references. Transaction similarity is computed using cosine similarity:

$$\text{sim}(x, y) = \frac{x \cdot y}{\|x\| \|y\|}$$

This measure captures the directional similarity between transaction representations and is robust to scale differences, making it suitable for cross-system alignment. Transactions exceeding a predefined similarity threshold are treated as candidate matches, while low-similarity pairs are flagged as discrepancies requiring further investigation.

Importantly, this approach supports decision support rather than automated resolution: similarity scores provide explainable evidence that guides reconciliation decisions while preserving audit trails and human oversight.

2.7.2 Interpretable Risk Scoring Models for Credit Decision-Support

Credit tracking and collection decisions within FAgentLLM are supported by interpretable risk scoring models designed to align with organizational credit policies and regulatory expectations. Rather than relying on black-box classifiers, the system employs weighted scoring formulations that mirror conventional financial risk assessment methodologies.

The overall risk score R associated with a customer or account is defined as:

$$R = \sum_{i=1}^n w_i f_i$$

where:

- f_i represents normalized financial indicators such as historical payment delays, outstanding exposure ratios, or dispute frequency,
- w_i denotes policy-defined weights reflecting the relative importance of each factor.

This formulation ensures that risk assessments remain transparent, adjustable, and auditable. LLM-powered agents interpret these scores contextually, using them to recommend actions such as reminder issuance, escalation, or review, rather than executing irreversible decisions. As a result, credit management remains a guided decision-support process grounded in both quantitative evidence and human judgment.

2.7.3 Cash Flow Projection Foundations for Liquidity Awareness

Effective cash management depends on timely awareness of short-term liquidity conditions. Within FAgentLLM, cash flow projections are employed as situational indicators that inform coordination across financial agents, rather than as autonomous optimization targets.

The projected cash balance for the subsequent period is estimated as:

$$\hat{C}_{t+1} = C_t + I_t - O_t$$

where:

- C_t denotes the current cash balance,
- I_t represents expected inflows,
- O_t represents expected outflows.

These projections provide a forward-looking signal that enables agents to reason about liquidity constraints when evaluating actions such as invoice approvals, budget commitments, or credit extensions. Crucially, cash projections are interpreted within context and subject to human validation, reinforcing their role as decision-support inputs rather than automated control mechanisms.

2.7.4 Role of Mathematical Foundations within the FAgentLLM Architecture

Collectively, these mathematical and technical foundations provide the executional substrate upon which LLM-powered agents operate. They do not replace judgment; rather, they structure it. By supplying interpretable signals—similarity scores, risk indicators, and liquidity projections—these models enable agents to reason about financial conditions while preserving traceability and governance.

The integration of lightweight analytical models with LLM-based contextual reasoning and multi-agent coordination allows FAgentLLM to restore systemic awareness to financial automation. In doing so, the architecture advances beyond task-level automation toward an intelligent, decision-support ecosystem that respects both the complexity and the accountability requirements of enterprise finance.

III REVIEW OF EXISTING APPROACHES

Financial automation has achieved remarkable domain-specific sophistication. Invoice processing systems extract data from unstructured documents with high accuracy. Budgeting platforms forecast expenditures using advanced analytics. Reconciliation tools identify discrepancies across millions of transactions. Credit management systems track receivables and automate collection workflows. Cash management platforms monitor liquidity in real time.

Yet as established in Section 2.1, operational efficiency within domains does not constitute systemic financial intelligence. An organization can process invoices flawlessly while remaining blind to liquidity implications. It can forecast budgets accurately while ignoring downstream credit exposure. It can reconcile transactions precisely while missing the structural patterns that signal process failure. Speed and accuracy are necessary but insufficient: financial intelligence requires coordination across interdependent domains.

This section systematically reviews existing intelligent automation approaches, progressing from domain-specific solutions (Sections 3.1–3.5) to integrated and multi-agent systems that attempt cross-domain coordination (Section 3.6). The five foundational domain: invoice management, budget management, reconciliation, credit tracking and collection, and cash management are first examined individually to establish their current capabilities and architectural boundaries. The review then turns to approaches that explicitly address integration, evaluating whether they achieve genuine systemic coordination or merely orchestrate independent modules.

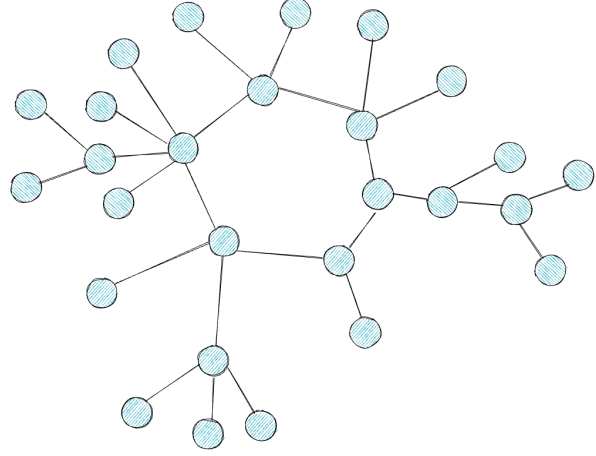


Figure 4 Conceptual system-level view of financial automation, illustrating how domain-specific approaches reviewed in Sections 3.1–3.5 and integration-oriented approaches in Section 3.6 collectively construct an interconnected network of interdependent processes

3.1 Intelligent Invoice Processing and Approval Automation

Authors / Year / Venue	Title	Focus & Research Question	Core Technique	Extraction Accuracy	Compliance Support	Key Findings	Limitations & Gaps	Relevance to Agent 1
Krieger et al. (2023) <i>Intelligent Systems with Applications</i>	<i>Automated Invoice Processing: ML-Based Information Extraction for Long-Tail Suppliers</i>	How can machine learning improve invoice information extraction for heterogeneous, low-frequency (“long-tail”) suppliers?	Supervised ML-based document information extraction for invoice field recognition and classification	High accuracy for structured fields; performance degrades on unseen or highly unstructured layouts	Indirect support via standardized outputs; compliance rules not explicitly modeled	ML approaches outperform rule-based systems for long-tail suppliers; improved scalability and robustness	No approval workflow modeling; no compliance reasoning; lacks semantic understanding; no LLMs or agent coordination	Foundational for Agent 1: establishes strong extraction baseline while exposing the need for reasoning, approval, and compliance-aware agents
Nie et al. (2024) <i>ACAI / IEEE</i>	<i>Towards a Conversational Invoice Issuance LLM-Based Agent</i>	How can LLMs automate invoice issuance in tax administration while handling regulatory complexity and incomplete user input?	LLM-based conversational agent with hierarchical memory (long-term user data, short-term context, external DB); supervised fine-tuning using ChatML	89.96/100 accuracy; 99.6% of human performance; 58× faster processing	Explicit compliance module using dual auditing (keyword filtering + RoBERTa classifier) achieving 93% recall	Conversational refinement improves accuracy; open-source LLM matches commercial models; memory significantly enhances performance	Limited to Chinese text-based invoices; no PDF/image processing; no scalability analysis for enterprise-scale workflows	Highly relevant: demonstrates LLM reasoning and compliance validation, but lacks multimodal extraction and enterprise workflow integration
Loukil et al. (2024) <i>FLLM 2024 / IEEE</i>	<i>LLM-Centric Pipeline for Information Extraction from Invoices</i>	Can OCR + prompt-engineered LLMs extract structured invoice data without fine-tuning while addressing OCR’s lack of semantic understanding?	Two-stage pipeline: OCR (Tesseract/DocTR) → prompt-engineered LLMs (Llama3/Mistral) → structured JSON with Pydantic validation	85–90% accuracy; F1-score up to 0.95; 2.18–3.43 sec per invoice	No explicit compliance reasoning; schema validation only	Open-source OCR + small LLMs achieve competitive accuracy without fine-tuning; cost-effective extraction	Dataset-dependent performance; no explanation of model selection; no handwritten invoices; no business-level error analysis	Relevant for Agent 1: supports LLM-assisted extraction, but highlights lack of compliance reasoning and adaptive model selection
Deepa et al. (2024) <i>IEEE ADICS</i>	<i>An Intelligent Invoice Processing System Using Tesseract OCR</i>	How can open-source OCR automate invoice text extraction from scanned and PDF invoices?	Tesseract OCR with CNN feature extraction, LSTM for context, template matching	80–90% accuracy (avg. 85%)	None beyond structured JSON output	Handles multiple layouts and formats; low-cost open-source solution	Template-dependent; manual intervention required for new layouts; no reasoning or validation	Low-level relevance: baseline OCR system illustrating limitations that motivate LLM-based reasoning agents

3.2 Automated Budgeting and Financial Planning Systems

Authors / Year / Venue	Title	Focus & Research Question	Methodology	Forecasting Capability	Scenario Analysis	Key Findings	Limitations & Gaps	Relevance to Agent 2
Tan et al. (2025), <i>Energy Strategy Reviews</i>	<i>IMPACCT: An Integrated Assessment Model for Policy and Financial Decision-Making in Energy Planning</i>	How can integrated assessment models support financial planning and budgeting for sustainable energy transitions?	Integrated Assessment Model combining seven open-source tools (MAED, OnSSET, OSeMOSYS, FlexTool, PathCalc, MINFin, FINPLAN) to link technical, social, and financial planning	Supports cost estimation and budget planning embedded within system-level projections (not pure time-series forecasting)	Enables exploration of alternative policy, investment, and socioeconomic scenarios affecting budgeting outcomes	Demonstrates improved coordination between technical planning and financial decision-making; supports evidence-based budget allocation	No real-time or adaptive budgeting; lacks automated budget revision mechanisms	Illustrates scenario-driven financial planning; informs high-level design of a Budget Management Agent
El Hamshary et al. (2022), <i>Tunnelling and Underground Space Technology</i>	<i>Integrated Maintenance Planning Approach to Optimize Budget Allocation for Subway Operating Systems</i>	How can maintenance planning and budget allocation be jointly optimized under limited budgets?	Risk-based multi-objective optimization framework using analytical models and Lagrangian multipliers	Focuses on optimization outcomes rather than explicit forecasting	Evaluates robustness of budget allocations under varying cost and risk scenarios	Integrated optimization significantly reduces failure risk compared to baseline methods	Deterministic and static; no data-driven forecasting or adaptive updates	Strong reference for constraint-based budget allocation logic within an automated agent
Huang (2024), <i>Journal of Organizational and End User Computing</i>	<i>Design and Implementation of an Intelligent Financial Management System Based on Enterprise Legal System</i>	How can deep learning improve financial planning, budget management, and revenue forecasting?	TCGNet model combining TCN and GNN optimized via Grey Wolf Optimization	Provides high-accuracy revenue forecasting to inform budgeting decisions	Implicit what-if analysis through varying promotional and financial inputs	Improves revenue prediction accuracy and supports planning decisions	No end-to-end automated budgeting workflow; limited discussion of real-time adaptation	Suitable foundation for integrating AI-based forecasting into a Budget Management Agent
Wang & Zareeihemat (2025), <i>Expert Systems with Applications</i>	<i>Multi-Channel Advertising Budget Allocation Using Q-Learning and Artificial Bee Colony</i>	How can reinforcement learning optimize long-term advertising budget allocation?	Q-learning enhanced with mutual learning-based Artificial Bee Colony and cost discretization	Implicit forecasting via sequential reinforcement learning	Explores channel-specific budget allocation strategies under varying constraints	Achieves strong optimization performance across multiple metrics	Narrow domain (advertising); lacks enterprise-wide budgeting context or explainability	Relevant for adaptive budget allocation mechanisms within a broader agent
Martseniuk et al. (2025), <i>Computers & Security</i>	<i>Cost Observability as a Security Control in Multi-Cloud Environments</i>	How can unified cost observability improve budget control and forecasting across clouds?	Unified cost lake architecture with regression and ensemble forecasting models	Explicit cost forecasting with improved RMSE, MAE, and R ² over native tools	Evaluates overspending risks under different budget thresholds	Enhances transparency and early detection of budget deviations	No agent-based decision automation or LLM reasoning	Supports cost-awareness layer for a Budget Management Agent

3.3 Reconciliation Automation and Anomaly Detection

Authors / Year / Venue	Title	Focus & Research Question	Matching Strategy	Anomaly Detection Approach	Accuracy / Performance	Key Findings	Limitations & Gaps	Relevance to Reconciliation Agent (Agent 3)
Yıldız et al. (2022), IEEE	<i>Anomaly Detection in Financial Data Using Deep Learning: A Comparative Analysis</i>	Which deep learning models best detect anomalous events in financial time series?	Indirect matching via next-step prediction of expected values in time series	Forecasting-based anomaly detection using reconstruction error (threshold: 3σ); LSTM, GRU, 1D-CNN compared	GRU achieved lowest error (e.g., Tesla MSE = 9.63); 1D-CNN performed worst	GRU outperforms LSTM and CNN for stable financial time series; hyperparameter tuning is critical	Performance degrades under extreme volatility; no handling of concept drift; not integrated into operational reconciliation workflows	Supports Agent 3 by enabling automated discrepancy detection via prediction error while maintaining traceability
Ramaswamy et al. (2025), IEEE	<i>Deep Learning-Based Anomaly Detection for Real-Time Financial Transactions in Cloud Environments</i>	How can real-time anomaly detection scale in distributed, high-volume transaction streams?	Pattern matching against learned normal transaction embeddings	Hybrid model: Autoencoder (feature extraction) + ANN (classification); MFFO used for load balancing	Accuracy: 99.8%; Precision: 98.58%; FPR = 0.0000	Achieves near-perfect detection with reduced latency and memory usage	Trained on a specific dataset; limited generalizability; lacks explainability (XAI)	Demonstrates scalable, real-time anomaly detection suitable for high-volume reconciliation pipelines
Hilal et al. (2022), Expert Systems with Applications	<i>Financial Fraud: A Review of Anomaly Detection Techniques and Recent Advances</i>	What anomaly detection techniques are effective for financial fraud under limited labeling?	Clustering-based matching (e.g., k-means, FCM) using distance measures	Categorizes anomalies as point, contextual, or collective; emphasizes unsupervised and semi-supervised methods	Random Forest: 96.2%; MLP + Whale Optimization: 98.04% F-score; SOM near-perfect in some settings	Strong shift toward unsupervised and deep learning methods due to label scarcity	Sparse public datasets; concept drift; focus skewed toward credit card fraud	Provides theoretical grounding for clustering-based reconciliation and rare discrepancy detection
Bakumenko & Elragal (2022), Systems	<i>Detecting Anomalies in Financial Data Using Machine Learning Algorithms</i>	How can anomalous journal entries be detected in general ledger data?	Feature-based matching using PCA-reduced journal entry vectors	Supervised (RF, SVM) and unsupervised (Isolation Forest, Autoencoder) anomaly detection	RF recall = 0.9925; Isolation Forest detected 100% of anomalies with 9.38% sampling	Tree-based and unsupervised models outperform others; dimensionality reduction critical	Highly imbalanced data; no sequence modeling; limited transaction depth	Directly applicable to GL reconciliation and audit-support automation
Manian & Kayal (2025), IIMB Management Review	<i>Detecting and Forecasting Financial Bubbles Using ML Models</i>	Can ML models detect market anomalies earlier than statistical methods?	Matching against historical “bubble” vs “non-bubble” regimes (GSADF test)	Right-tailed unit root tests + ML classifiers (RF, GB, ANN)	RF & GB accuracy = 1.00; ANN = 0.945	Ensemble models capture complex financial deviations effectively	Relies on historical macro data; limited real-time validation	Demonstrates regime-level anomaly detection for systemic reconciliation signals

*Bakumenko et
al. (2025),
IEEE Access*

*Advancing
Anomaly
Detection: Non-
Semantic
Financial Data
Encoding with
LLMs*

Can LLM
embeddings
improve anomaly
detection in
heterogeneous
financial records?

Group-level
matching using
SBERT
embeddings of
transaction sets

SBERT + ML
classifiers (NN,
XGBoost) detect
unusual transaction
patterns

Macro recall $\approx$
0.95; outperforms
one-hot and
autoencoder
baselines

Dense embeddings
reduce sparsity and
improve cross-
format
reconciliation

Synthetic anomalies;
limited
interpretability; no
numeric reasoning

Highly aligned with
Agent 3 for
standardizing
heterogeneous
records and
improving
reconciliation
accuracy

3.4 Credit Risk Assessment and Collection Automation

Authors / Year / Journal	Title	Focus & Research Question	Scoring Method	Extraction Accuracy	Collection Strategy	Key Findings	Limitations & Gaps	Notes and Relevance to Agent 4
Peng, Q.; Zhu, Y.; Wang, G.-J. (2025) <i>International Review of Financial Analysis</i>	<i>SCF credit risk assessment with limited labeled data using label propagation algorithm and complex network approaches</i>	How can enterprise credit risk in Supply Chain Finance (SCF) be accurately assessed when labeled data are scarce?	Semi-supervised Label Propagation Algorithm (LPA) and DGCN-optimized LPA, leveraging financial indicators and inter-firm complex network structures	Not applicable. The study evaluates risk classification performance rather than document or invoice-level extraction accuracy	Implicit credit monitoring and prioritization based on network centrality and risk propagation patterns; no explicit debt collection automation	1. LPA-based methods outperform traditional supervised and semi-supervised learning approaches under limited labeled data. 2. DGCN improves predictive performance by capturing both local and global network dependencies	Focuses on enterprise-level credit risk classification rather than invoice-level delinquency, payment behavior, or recovery actions	Provides a strong quantitative foundation for Agent 4's credit risk scoring module, particularly in data-scarce environments. The graph-based inference mechanism can be adapted to infer latent credit risk signals prior to triggering collection actions
Khalil, M. A.; Hadid, M.; Padmanabhan, R.; Elomri, A.; Kerbache, L. (2025) <i>Decision Analytics Journal</i>	<i>An integrated Artificial Intelligence and optimization model for operational efficiency and risk reduction in Letter of Credit examination process</i>	How can AI and optimization mitigate operational inefficiencies and quantifiable risks in hybrid (AI + human) financial document processing?	Risk scoring using a weighted discrepancy criticality scale (1–10) based on financial impact, checker error probability, and transaction value	Explicit OCR accuracy is not reported; however, the framework achieved a 68.3% reduction in operational risk, with up to 96.65% reduction in controlled scenarios	AI-based document processing (OCR, NLP, ML) combined with Integer Linear Programming (ILP) to optimally allocate verification tasks under capacity constraints	1. Achieved utilization rates between 34% and 73%. 2. Identified a critical utilization threshold (~70%) beyond which operational risk increases sharply	Relies on deterministic parameters; evaluated within a single international banking network; does not explicitly address post-examination credit collection or recovery	Serves as a direct technical blueprint for Agent 4. The risk scoring formulation and ILP-based workflow optimization can be repurposed to automate collection prioritization, escalate high-risk accounts to senior collectors, and support human-in-the-loop decision making for high-value or legal credit actions

3.5 Cash Flow Forecasting and Liquidity Management

Authors / Year / Journal	Title	Focus & Research Question	Method	Performance Metrics	Dataset & Features	Key Findings	Limitations & Gaps	Notes and Relevance to Agent 5
Subiarti, S.; Saepudin, E. (2025) <i>IEEE International Conference on Artificial Intelligence for Learning and Optimization (ICoAILO)</i>	<i>Long Short-Term Memory (LSTM) Neural Network for Predicting Free Cash Flow (FCF) in Financial Sector Companies Listed on the Indonesia Stock Exchange (IDX)</i>	Can LSTM combined with Greedy Forward Selection accurately predict Free Cash Flow (FCF) for financial sector firms?	LSTM neural network with Greedy Forward Selection and SHAP for explainability	MAPE: 16.07%; RMSE: 0.1187; MAE: 0.0823	252 samples from four IDX-listed financial firms (2008–2024); features included lagged FCF, EPS, ROA, ROE	Optimal performance with lagged FCF features; LSTM outperformed linear baselines	Small sample size; limited to Indonesian financial sector	Validates core forecasting model for Agent 5's liquidity simulator. Supports LSTM-based cash flow prediction and SHAP explainability as decision support inputs
Tiwari, A.; et al. (2025) <i>World Skills Conference on Universal Data Analytics and Sciences (WorldSUAS)</i>	<i>Enhancing Cash Flow Management with ML Approaches and Cloud-Based Tools</i>	How can ML and cloud infrastructure improve cash flow forecasting and real-time liquidity decisions?	Hybrid ML-cloud architecture: LSTM + XGBoost with integrated ERP/CRM/POS data pipeline	Forecast Accuracy: 93.4%; Optimization Efficiency: 91.2%; Error Reduction: 71.5%; Response Time: 85.3%	Enterprise datasets capturing historical revenue, expenses, and operational cash flows	ML-cloud system outperformed rule-based tools; strong adaptability to volatility	Cloud data privacy and latency concerns; lack of autonomous reinforcement learning	Provides robust forecasting and integration architecture for Agent 5. Supports centralized financial data ingestion and high-accuracy, real-time forecasting
Mutian, Y.; Lin, Q.; Zhang, L.; et al. (2025) <i>Franklin Open (Elsevier)</i>	<i>LLM-guided semantic feature selection for interpretable financial market forecasting in low-resource financial markets</i>	Can LLMs perform zero-shot semantic feature selection to improve forecast accuracy and interpretability in low-resource markets?	LLM-guided Retrieval-Augmented Feature Selection (RAFS) with XGBoost prediction	RMSE: 12.82; MSE: 164.38; R ² : 0.75	Weekly macro-financial data with 15 candidate features (e.g., index, commodities, exchange rates)	LLM-guided features outperform traditional selection in noisy, low-resource environments	Prompt sensitivity; reliance on proprietary LLMs; temporal dependencies handled downstream	Validates LLM-assisted feature intelligence layer for Agent 5. Enhances explainable feature selection and supports forecasting in environments with limited structured data
De Oliveira, E. Lucas, P.; Guimarães, F. G.; Mendes, E. M. (2025) <i>Neurocomputing, 656, 131569</i>	<i>Automated machine learning based on decomposition, causality and evolutionary multitask optimization for time series forecasting</i>	How can decomposition, causality analysis, and evolutionary multitask optimization improve automated time series forecasting performance?	AutoDCE-TS AutoML framework integrating time series decomposition, causality-based feature selection, and evolutionary multitask optimization for model search	Standard forecasting error metrics (e.g., RMSE, MAE) with consistent performance improvements reported across multiple benchmarks	18 diverse time series datasets covering multiple domains (not limited to finance)	AutoDCE-TS demonstrates robust, general-purpose forecasting performance competitive with 10 comparative methods through automated feature extraction, model selection, and	Not evaluated specifically on cash flow or corporate finance datasets; general-purpose design requires financial adaptation	Offers a powerful AutoML forecasting engine for Agent 5 capable of automating model selection and optimization. With domain-specific adaptation, it can strengthen automated liquidity forecast generation

<i>Zhao, H.; Liu, Z.; Wu, Z.; et al. (2025) Survey Paper</i>	<i>Revolutionizing Finance with LLMs: An Overview of Applications and Insights</i>	How effectively can LLMs perform diverse financial tasks using natural language instructions?	Comprehensive survey and empirical evaluation of GPT-4 under zero- and one-shot prompting	Accuracy, EntityF1, and Exact Match scores across six financial datasets	Multiple datasets including sentiment, entity recognition, financial QA, and market signal prediction	optimization GPT-4 demonstrates strong reasoning, instruction following, and financial text processing without extensive retraining	LLMs require integration with quantitative models for numeric forecasting and optimization	and reduce manual model engineering Provides theoretical and empirical justification for incorporating LLM reasoning in Agent 5. Enables interpretation of market signals and guide proactive liquidity decisions
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3.6 Multi-Agent and Integrated Financial Systems

Authors / Year / Journal	Title	Focus & Research Question	Agent Design	Coordination Mechanism	LLM Use	Key Findings	Limitations & Gaps	Notes and Relevance to Current Study
Huang et al. (2025) <i>International Conference on Data Science, Agents & Artificial Intelligence (ICDSAAI)</i>	<i>Scalable Invoice Reconciliation for SMEs: Edge-Deployed LLMs in Multi-Agent Systems</i>	How can invoice reconciliation be automated using edge-deployed LLM agents while ensuring cost efficiency, privacy, and scalability for SMEs?	Role-specialized agents (Data Engineering, Email, Reconciliation) with explicit task decomposition and tool usage	Graph-based orchestration (LangGraph) enabling sequential and conditional agent interactions	Evaluates proprietary (GPT-4o, GPT-4o-mini) and open-source models (Qwen2.5, Functionary V3.1, LLaMA-3.2-Vision) deployed locally via Ollama on edge GPUs	Qwen2.5-32B achieved 99.9% success rate, outperforming proprietary models; Functionary-small showed optimal performance-to-power ratio for constrained hardware	LLM hallucinations can cause OCR or SQL generation failures; VRAM constraints limit scalability of very large models; real-time ERP integration remains unresolved	Directly validates the architectural core of FAgentLLM. Confirms the feasibility of edge-deployed, role-based LLM agents for invoice workflows, supports LLMs as an intelligence layer for structured extraction, and aligns with FAgentLLM's privacy-preserving, cost-efficient, multi-agent orchestration
Bangun et al. (2025) <i>3rd International Conference on Advancement in Computation & Computer Technologies (InCACCT)</i>	<i>Multi-Agent Conversational System for Automated Stock Market Analysis and Decision Support Toward Inclusive Economic Growth</i>	Can hybrid multi-agent financial reasoning (fundamental, technical, macro) improve accuracy and explainability under real-time constraints?	Five agents: Intent, Fundamental (P/E, FCF), Technical (RSI, MACD), Macroeconomic, and Summarizer	CrewAI-based coordination, using YAML-defined workflows, role separation, and shared contextual state	GPT-based and LLaMA-based LLMs for intent detection, reasoning, and conversational synthesis	Hybrid approach outperformed single-dimension analysis; Fundamental Agent achieved 75% directional accuracy	Limited handling of sentiment shocks and exogenous events; lack of comparative analysis across agent architectures	Validates multi-agent financial reasoning and structured coordination. Directly informs FAgentLLM's Cash Management and Credit Agents through FCF-based analysis and supports explainable synthesis via summarization agents
Fazlija et al. (2025) <i>Journal of Behavioral and Experimental Finance</i>	<i>Reasoning with Financial Regulatory Texts via Large Language Models</i>	How effectively can LLMs reason over complex financial regulations (e.g., Basel III) to derive correct risk-weighted outcomes?	Single-LLM reasoning architecture; no explicit agent role separation	Implicit coordination via Chain-of-Thought (CoT) and Tree-of-Thought (ToT) reasoning strategies	GPT-4, GPT-4o, and Claude evaluated under zero-shot and few-shot prompting	Few-shot CoT significantly improves regulatory reasoning accuracy; LLMs can reliably extract and apply regulatory logic	No multi-agent design; limited to regulatory text reasoning; not evaluated in operational financial workflows	Supports FAgentLLM's intelligence-layer philosophy, particularly for compliance-aware agents (e.g., Credit or Risk Agents), but does not address agent coordination or automation

<i>Gupta et al. (2025) 9th International Conference on Computational System and Information Technology for Sustainable Solutions (CSITSS)</i>	<i>A Preliminary Fundamental Financial Analysis Framework Using Structured LLM Prompting</i>	Does providing pre-calculated financial metrics to LLMs outperform raw financial statement inputs?	Modular pipeline computing 15 financial metrics across liquidity, solvency, profitability, efficiency, and risk	Linear, modular workflow from CSV ingestion to LLM-driven interpretation	GPT-4, Claude, and Gemini used strictly as interpreters of structured inputs	Structured inputs yielded 73% higher relevance, 81% better risk detection, and 65% improved comparative accuracy	Dependency on fixed CSV schemas; reliance on external APIs; lacks dynamic industry customization	Provides strong empirical evidence for FAgentLLM's design choice: LLM agents perform best when consuming structured, pre-computed financial indicators—directly strengthening Credit Tracking and Cash Management Agents
<i>Yang et al. (2024) arXiv</i>	<i>LLM-based Multi-Agent Systems: Techniques and Business Perspectives</i>	How can LLM-based agents evolve into collaborative artificial collective intelligence while preserving privacy and scalability?	Modular LaMAS architecture with interaction wrappers, memory layers, cognitive reasoning, and tool integration	Identifies Star, Ring, Graph, and Bus coordination patterns; proposes a Core Protocol Framework	LLMs act as the cognitive core using prompt engineering, few-shot learning, and MARL	Multi-agent systems improve robustness, specialization, and fault tolerance; Decentralized Star Architecture enhances privacy	Communication overhead, semantic privacy leakage, and scalability challenges persist	Theoretical and architectural justification for FAgentLLM. The Credit Allocation Protocol and decentralized coordination offer a direct blueprint for secure interaction among Invoice, Budget, Credit, and Cash Agents
<i>Fatemi & Hu (2024) Proceedings of the 5th ACM International Conference on AI in Finance (ICAIF '24)</i>	<i>FinVision: A Multi-Agent Framework for Stock Market Prediction</i>	Can LLM-based agents integrate multimodal financial data while improving interpretability and performance over RL models?	Four agents: News Summarizer, Technical Analyst (charts), Reflection Agent, and Prediction Agent	LangGraph-based directed graph, with feedback loops via a Reward and Reflection mechanism	GPT-4o-mini for multimodal reasoning; o1-mini for advanced prediction; CoT prompting for transparency	Outperformed PPO, DQN, and traditional indicators; reflection agent critical for adaptive learning	Underperformed FinAgent benchmark due to shorter training duration; RL integration remains limited	Provides a technical precedent for interconnected financial agent ecosystems. For FAgentLLM, it supports reflective feedback loops and offers concrete strategies for cash reserve and liquidity compliance in Agent 5

3.7 Beyond Domain-Level Optimization: Limits of Current Financial Automation

Taken collectively, the reviewed literature demonstrates substantial progress in automating individual financial functions, with notable gains in accuracy, scalability, and computational efficiency across invoice processing, budgeting, reconciliation, credit risk assessment, and cash flow forecasting. Within each domain, state-of-the-art techniques ranging from deep learning and ensemble models to optimization-based frameworks and LLM-assisted reasoning have significantly outperformed traditional rule-based systems. These advances confirm that domain-level automation is no longer a technical bottleneck in modern financial systems.

However, a cross-sectional examination reveals a persistent structural limitation: most approaches remain functionally siloed. While models within each domain optimize local objectives such as extraction accuracy, forecasting error, or anomaly detection, they rarely account for downstream financial dependencies or upstream causal feedback. As a result, improvements in one function (e.g., invoice extraction or budget forecasting) do not systematically translate into improved enterprise-level financial intelligence. Integration efforts, where present, typically focus on data pipelines or workflow orchestration rather than shared reasoning, cross-domain state awareness, or coordinated decision-making.

Furthermore, methodological differences across domains expose deeper inconsistencies in how intelligence is operationalized. Forecasting-oriented approaches prioritize predictive accuracy but often lack interpretability and decision linkage. Optimization-based systems deliver strong prescriptive outcomes yet rely on static assumptions or predefined constraints. LLM-based methods excel at semantic understanding and instruction following but remain dependent on external quantitative models for numerical reasoning and control. These differences highlight not only complementary strengths, but also unresolved trade-offs between accuracy, explainability, adaptability, and autonomy.

Most critically, the literature reveals a gap between automation and systemic intelligence. Few studies explicitly address how financial decisions propagate across domains, how conflicting objectives are reconciled, or how agents can collaboratively reason under uncertainty. The absence of unified coordination mechanisms, feedback loops, and shared semantic representations limits the ability of existing systems to move from efficient task execution to coherent, organization-wide financial decision support.

In light of these observations, the next section conducts a critical analysis of the reviewed approaches. It systematically evaluates their strengths and weaknesses, contrasts their underlying assumptions and design philosophies, and identifies unresolved research gaps that motivate the need for an integrated, multi-agent financial intelligence framework.

IV CRITICAL ANALYSIS

4.1 Overview

Building on the systematic review presented in Section 3, this section critically evaluates the most impactful contributions in each financial domain. The analysis focuses on strengths and weaknesses of methods, contrasts differing methodological paradigms, and identifies structural research gaps that current automation approaches have yet to overcome. These gaps directly motivate the integrated, multi-agent architecture underlying FAgentLLM.

4.2 Comparative Evaluation of Best-in-Class Approaches

4.2.1 Invoice Management

Loukil et al. (2024) is selected as the strongest contribution within invoice management research. The system’s two-stage architecture—optical text extraction via OCR (Tesseract/DocTR) followed by structured data generation using prompt-engineered large language models (LLMs)—achieves approximately 90% extraction accuracy across diverse invoice formats while maintaining practical processing times (2.18–3.43 seconds per document). Crucially, the integration of schema validation (e.g., Pydantic) ensures extracted entities conform to expected financial structures, enhancing both precision and reliability.

When contrasted with alternatives, *Krieger et al. (2023)* achieve high extraction accuracy but lack semantic reasoning and cannot generalize to unstructured layouts. *Nie et al. (2024)* demonstrate substantial throughput improvements but are constrained to tax administration in Chinese contexts, limiting broader applicability. *Deepa et al. (2024)* rely on template-dependent methods that require manual adaptation for vendor-specific variations, undermining scalability.

Despite its strengths, Loukil et al.’s approach does not support multi-modal inputs (e.g., handwriting), lacks dynamic cost-benefit analyses for model selection, and operates in functional isolation without considering downstream workflows such as approval logic or compliance escalation.

4.2.2 Budget Management

The work of *Huang (2024)* is identified as the principal contribution in budgeting systems due to its integration of temporal convolutional networks (TCN), graph neural networks (GNN), and meta-heuristic optimization (Grey Wolf Optimization) for revenue forecasting linked to budget allocation. The model delivers high predictive performance (reported accuracy ~99.8%) and connects forecasting outputs to decision support mechanisms, distinguishing it from predictive tasks treated in isolation.

While *Tan et al. (2025)* integrate multiple tools for policy and financial scenario analysis, their focus is not on automated budgeting or real-time adaptation. *El Hamshary et al. (2022)* and *Wang & Zareehemat (2025)* employ optimization frameworks with limited forecasting integration, and *Martseniuk et al. (2025)* enhance cost observability without adaptive decision automation.

Huang’s model advances the field by addressing both temporal patterns and relational dependencies, but it does not articulate automated workflows for alerts, approval triggers, or interactions with liquidity or reconciliation subsystems. Additionally, model explainability remains limited, constraining interpretability for stakeholders.

4.2.3 Reconciliation Automation

Bakumenko et al. (2025) represents the strongest reconciliation contribution, pioneering the use of large language model-derived embeddings (e.g., SBERT) to encode heterogeneous, non-semantic financial record elements. By mapping high-cardinality categorical fields into dense feature spaces with high variance retention and then classifying anomalies via standard machine learning models, this approach surpasses baseline encoding techniques and traditional autoencoder models in recall and scalability.

Competing methods such as *Yıldız et al. (2022)* rely on traditional sequential modeling, and *Ramaswamy et al. (2025)* demonstrate high detection rates but lack interpretability and generalizability. Other contributions focus on fraud detection contexts not directly translatable to reconciliation (*Hilal et al., 2022*) or utilize statistical tests without modern feature representation (*Manian & Kayal, 2025*).

Nevertheless, the reliance on synthetic anomalies and lack of explainable outputs remains a limitation. The approach also omits integration with either upstream (process causality) or downstream (decision support) modules.

4.2.4 Credit Tracking and Collection

Within credit domain research, *Peng et al. (2025)* is chosen for its semi-supervised graph-based approach, integrating label propagation with deep graph convolutional networks (DGCN) to model inter-firm credit risk propagation in data-sparse environments. This enables credit scoring even when labeled data are limited—an acknowledged challenge in enterprise finance.

Comparatively, *Khalil et al. (2025)* apply AI-augmented optimization to Letters of Credit workflows, yielding substantial operational risk reductions but relying on deterministic parameters and lacking integration with broader credit tracking or real-time collection automation.

While Peng et al.'s method excels in cold-start risk scoring, it neither models invoice-level payment behavior nor supports automated collection or debt aging strategies. The model's static nature also precludes real-time updates based on operational feedback from other financial functions.

4.2.5 Cash Flow Forecasting

The contribution of *Tiwari et al. (2025)* is identified as the most comprehensive within cash flow forecasting. The architecture integrates LSTM networks, XGBoost models, and cloud-based data ingestion pipelines to deliver scalable, real-time forecasting with robust performance metrics (e.g., ~93.4% accuracy). Unlike isolated predictive tasks, the framework includes decision support mechanisms that align forecasting outputs with liquidity planning.

By contrast, *Subiarti & Saepudin (2025)* provide strong localized forecasting results but are constrained by dataset size and feature scope, while *Mutian et al. (2025)* and *Zhao et al. (2025)* highlight semantic and reasoning capabilities without offering integrated operational pipelines.

Tiwari et al.'s framework nonetheless lacks adaptive reinforcement learning components and does not reason about cross-domain dependencies (e.g., invoice settlement timing or budget constraints), limiting its systemic utility.

4.3 Synthesis of Methodological Strengths

Across domains, three recurring methodological strengths emerge:

1. **Deep Learning for Complex Financial Patterns:** Neural architectures—LLM encoders, sequence models, graph networks—consistently outperform rule-based or statistical baselines in representation and prediction tasks.
2. **Hybrid Architectures as Best Practice:** Combinations of neural models, pre-trained embeddings, and optimization components achieve superior performance compared to monolithic approaches, indicating that financial complexity requires diverse inductive biases.
3. **Transfer Learning for Generalization:** Pre-trained models (LLMs, SBERT) demonstrate flexibility in handling novel formats and evolving data structures, suggesting that transfer learning is integral to scalable financial automation.

4.4 Critical Methodological Weaknesses

Despite these strengths, current approaches share five fundamental limitations:

1. **Domain Isolation and Lack of Causal Modeling:** All reviewed systems optimize local performance without modeling causal dependencies between financial decisions. High extraction accuracy or forecast precision does not ensure coherent enterprise financial management when decision effects propagate across domains.
2. **Explainability Deficits in High-Stakes Contexts:** Black-box solutions undermine accountability and auditability. Predictive accuracy is insufficient where interpretive justification is required for regulatory compliance or strategic planning.
3. **Static Workflow Assumptions:** The reliance on historical data and fixed workflows reduces resilience to novel formats, regulatory shifts, or systemic disruptions, necessitating expensive retraining.

4. **Absence of Semantic Cross-Domain Coordination:** While some frameworks orchestrate modular agents, none integrate semantic reasoning about inter-domain implications (e.g., how a reconciliation discrepancy affects credit risk or liquidity thresholds).
5. **Limited Operational Validation:** Validations on controlled datasets fail to test systems under real-world complexities—format variability, incomplete records, contradictory information, or exogenous shocks.

V SUMMARY AND CONNECTION TO FAgentLLM

The critical evaluation presented in Section 4 highlights a fundamental insight: although contemporary financial automation systems demonstrate impressive local performance within discrete domains, they remain architecturally siloed and fail to deliver the integrated intelligence necessary for enterprise-level financial decision-making. This shortfall is not a minor limitation of model tuning or workflow design; it reflects a structural design paradigm that prioritizes task-specific optimization, predictive accuracy, or domain-centric efficiency over cross-domain causal reasoning, semantic understanding, and adaptive coordination.

The literature review identifies several persistent structural gaps: the absence of causal integration across financial functions, the underutilization of LLMs as cognitive engines, insufficient context-aware adaptability, lack of coherent multi-agent orchestration, and limited validation under realistic operational complexity. Collectively, these deficits reveal that incremental improvements in isolated models are insufficient—no existing system achieves simultaneous explainability, adaptive reasoning, and operational resilience, leaving enterprise financial intelligence unrealized.

FAgentLLM directly addresses these limitations through a paradigm-shifting, multi-agent architecture. It orchestrates five specialized agents—Invoice Management, Budget Management, Reconciliation, Credit Tracking, and Cash Management—operating over a shared, causally-informed financial state that encodes interdependencies across domains. By elevating LLMs from auxiliary analytical components to core cognitive infrastructure, FAgentLLM empowers each agent to generate transparent, natural language reasoning traces, thereby ensuring auditability, regulatory compliance, and actionable accountability for financial decision-makers.

5.1 Linking Literature to FAgentLLM

FAgentLLM leverages the strongest methodological insights from the literature while extending beyond the limitations identified. Like the leading studies, it employs a distributed multi-agent architecture. However, it surpasses prior work by integrating LLMs as an “intelligence layer,” transforming semantic extraction (Loukil et al., 2024) and embedding-based anomaly detection (Bakumenko et al., 2025) into a cohesive, enterprise-level reasoning engine. Unlike isolated approaches, FAgentLLM enables agents to reason about causal implications across domains: invoice approvals are evaluated in the context of cash flow and budgetary constraints, credit scores reflect real-time payment patterns, and reconciliation insights dynamically inform collection strategies.

5.2 Transforming Current Limitations into Systemic Capabilities

FAgentLLM enacts a holistic paradigm shift by addressing the structural deficits pervasive in current research:

- **From Isolation to Causal Intelligence:** Each agent interprets its domain actions in relation to shared financial state. For instance, the Invoice Agent evaluates cash availability before approving payments, ensuring that local decisions support enterprise-wide liquidity and risk objectives.
- **From Black Boxes to Explainable AI:** By using LLMs as cognitive infrastructure, agents produce interpretable reasoning traces—professionals can understand why a budget was deferred, a risk score was raised, or a cash forecast was revised. This transparency strengthens accountability and facilitates regulatory compliance.
- **From Static to Adaptive Workflows:** Agents adapt dynamically to new invoice formats, regulatory changes, or evolving market conditions using prompt engineering rather than retraining, ensuring resilience to operational novelty without sacrificing accuracy.
- **From Syntax to Semantic Coordination:** Through an event-driven architecture, agents articulate cross-domain implications. For example, when the Reconciliation Agent detects systematic payment anomalies, the Credit Tracking Agent adjusts risk assessments accordingly, and the Collection Agent adapts its strategy, ensuring context-aware coordination.

- **Resilience to Operational Complexity:**

FAgentLLM is explicitly designed to navigate the “messiness” of real-world finance: it resolves contradictory information through multi-agent deliberation, handles incomplete data gracefully, and maintains audit-ready reasoning trails, ensuring reliable decision-making under uncertainty and operational stress.

By integrating natural language reasoning with structured semantic representations, FAgentLLM transforms high-accuracy extraction, forecasting, and anomaly detection into holistic, enterprise-level financial intelligence. This architecture operationalizes the cross-domain causal interdependencies that existing literature overlooks, elevating financial automation from a set of high-performing silos to a systemic, explainable, and adaptive intelligence framework capable of navigating complexity, uncertainty, and novelty in real-world operations.

Yet a critical question remains: *which methodological choices will ensure these integrated capabilities genuinely support decision-making rather than remaining conceptual sophistication on paper?* The next section examines these choices spanning optimization, simulation, and data collection providing a bridge between theoretical promise and operational realization.

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